

Gas Absorption

PRE-LAB PLAN

Unit Operations Lab 6

4/8/2026

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1. Experimental Objective

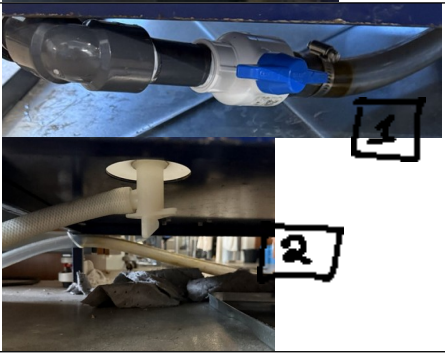
The first objective of the lab *Gas Absorption* is to investigate how water and gas flowrates affect the pressure drop and flooding of the gas absorption column. The second objective of the lab is to study how CO₂ is transferred into the water and to determine the overall mass transfer coefficient in both an open and closed setup. This is accomplished by using the UOP 7 gas absorption column and a titration setup to determine the concentration of CO₂ over time.

2. Experimental

This experiment is divided into three sections. Experiment 1 will investigate the pressure drop across the bed at varying gas and liquid flowrates. Experiment 2 will investigate the mass transfer in the column in a closed system and experiment 3 will investigate the mass transfer in an open system instead.

2.1 Apparatus

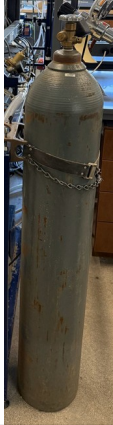
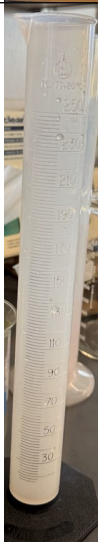
#	Name	Main/ Auxiliary	Description	Visual
1	Armfield Gas Absorption column (UOP7)	Main	Used to carry out experiments. Holds Raschig rings and gas/water flow.	

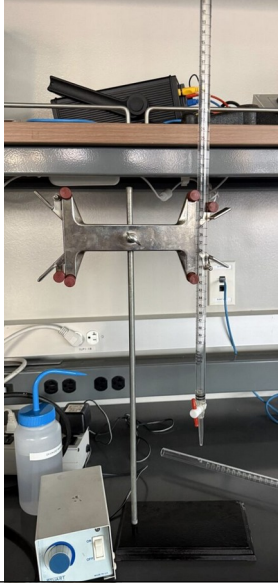
2	[1] Air flow meter [2] CO₂ flow meter [3] Water flow meter	Auxiliary	[1] Reads flow of compressed air coming from lab supply [2] Reads flow of CO₂ coming from cylinder [3] Reads flow of water coming from tank	
3	DI Water tank	Main	Holds DI water used within experiment	
4	Pump	Main	Used to initiate airflow	
5	Drain Valves [2] Tank [1] Column	Auxiliary	[2] Drains water from tank [1] Drains water from column	

6	Manometer	Auxiliary	Reads pressures from column	
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2.1.1 Materials

#	Name	Material/Supply	Description	Visual
1	Compressed Air	Supply	Used in experiment	

2	CO_2	Supply	CO_2 used throughout system	
3	Bucket	Material	Used for transporting Raschig rings/water	
4	Graduated Cylinder	Material	Used when finding void fraction	

5	Titration Setup	Material	Used to perform tests on experiment samples	
6	Raschig Rings	Material	Filled in column for performing flooding, absorption	

2.2 Procedure

Independent Variables: Gas flow Rate, CO₂ Flow rate, Liquid Flow Rate, Closed Loop vs. Open, Dry bed vs. Wet bed, sampling intervals

Dependent Variables: Pressure Drop. Flooding point, CO₂ concentration in liquid phase, CO₂ concentration in gas phase, Mass transfer performance

Part 1: Packed Bed Characterization

1. Fill a graduated cylinder with a known volume of packing material.
2. Slowly add water until all void spaces are filled.
3. Record the volume of water added.
4. Calculate the void fraction.

Pressure Drop Measurements

1. Close all drain valves and fill the liquid reservoir to about 3/4th full of deionized water and record the temperature.
2. Turn on the air supply Valve C2, and set the airflow to 20L/min. Let the air flow for 5 minutes to dry the packing.
3. Measure the pressure drop across the dry bed at different gas flow rates using the manometer.
4. Start the liquid flow by turning on the pump and opening valve C1 and C4
5. Set the water flow to 1.5L/min
6. Adjust the liquid level to remain above the sampling port and below the air inlet.
7. Reintroduce air and measure the pressure drop as gas flow increases until flooding occurs.
8. Repeat for additional liquid flow rates such as 3,6,8 L/min
9. Monitor the manometer to avoid overflow at high pressure drops.

Part 2: CO₂ Absorption and Desorption in a Closed system

1. Fill the reservoir again to about 3/4th of deionized water and record the temperature.
2. Collect an initial water sample and determine CO₂ concentration by titration.
3. Start the liquid circulation at 6 L/min and adjust the air flow to 20 L/min
4. Open the CO₂ cylinder and adjust the flow between 2-5 L/min
5. After about 5 minutes, collect 50 mL water samples periodically until CO₂ concentration stabilizes.
6. Measure the outlet gas CO₂ composition using the Hempl apparatus.

Desorption:

1. Turn off the CO₂ flow and reduce liquid flow to 1.5 L/min
2. Open the drain valve and adjust the liquid level.
3. Begin collecting water samples at short time intervals such as 1 minute
4. Analyze samples by titration to monitor CO₂ removal

Part 3: CO₂ Absorption with and open system

1. Refill the reservoir and measure the initial CO₂ concentration.
2. Start the liquid flow to 1.5 L/min and open the drain valve for continuous operation.
3. Maintain proper liquid level using drain valve control.
4. Set air flow to 20 L/min
5. Turn on the CO₂ flow rate between 2-6 L/min and turn on the heater block switch
6. Collect water samples every 30 seconds for the first 2 minutes, then every 1 minute for the next 5 minutes, then every 5 minutes until it reaches steady state.
7. Measure the outlet gas CO₂ concentration using the Hempl apparatus.

Data Analysis:

1. Plot the pressure drop vs. Gas flow rate at different liquid flow rates
 - a. Identifying the flooding points
2. Plot the CO₂ concentrations vs time for part 2 and part 3 of the experiment
3. Mass transfer coefficient calculation along with an interpretation of the data

2.3 Safety

Safety Analysis:

Hazard	Yes/No	If Yes, Description
Potential electrical hazards?	yes	Pump for process water and heating unit for compressed gas cylinder
Potential mechanical hazards?	yes	Pump is a rotating piece of equipment
Potential pressure hazards?	yes	CO ₂ gas bottle is at ~700 psia and process can reach pressures of 40 psia.
Potential temperature hazards?	yes	Heating element attached to gas bottle to offset the Joule–Thomson effect.
Hazardous/toxic chemicals involved	yes	CO ₂ is an asphyxiant
Gloves required?	yes	Wearing gloves is good lab safety practice
MSDS reviewed by all team members	yes	CO ₂ (124-38-9), Water(7732-18-5)
Process parameters		
Temperature (°C)	25	
Pressure (psia)	40	
Safety controls	Yes	

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I have read relevant background material for the Unit Operations Laboratory entitled: “Gas Absorption” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Alexis Guillen</u>	<u>4/8/26</u>
Signature of team member	Date
<u>Khoa Nguyen</u>	<u>4/8/26</u>
Signature of team member	Date
<u>Rishabh S. Patel</u>	<u>4/8/26</u>
Signature of team member	Date
<u>Kyle Hankosky</u>	<u>4/8/26</u>
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<u>Miguel Piza</u>	<u>4/8/26</u>
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